

INTEGRATION BY PARTS

If u and v are functions of x , by product rule:

$$\frac{d}{dx}(uv) = uv' + u'v$$

Re-arranging: $uv' = \frac{d}{dx}(uv) - u'v$

Integrating both sides w.r.t x , we have

$$\int uv' dx = uv - \int u'v dx \quad \leftarrow \text{Integration by parts formula:}$$

This is known as integration by parts formula and is very useful to integrate products of 2 functions such as:

$$\int x e^x dx, \int x \cos x dx, \int (x^2+1) \ln x dx, \text{ etc}$$

Example 1 (Evaluate:)

(a) $\int x e^{2x} dx$

(b) $\int x^2 \ln x dx$

(c) $\int e^{3x} \sin x dx$

Solution

(a) $\int x e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{2} \int e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C$

$$\begin{cases} \text{let } u = x & v' = e^{2x} \\ u' = 1 & v = \frac{1}{2} e^{2x} \end{cases}$$

(b) $\int x^2 \ln x dx = \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 dx = \frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 + C$

$$\begin{cases} \text{let } u = \ln x & v' = x^2 \\ u' = \frac{1}{x} & v = \frac{1}{3} x^3 \end{cases}$$

$$(c) \int e^{3x} \sin x \, dx = -e^{3x} \cos x + 3 \int e^{3x} \cos x \, dx$$

$$\left\{ \begin{array}{l} \text{let } u = e^{3x} \quad v' = \sin x \\ u' = 3e^{3x} \quad v = -\cos x \end{array} \right.$$

$$\left\{ \begin{array}{l} \text{let } u = e^{3x} \quad v' = \cos x \\ u' = 3e^{3x} \quad v = \sin x \end{array} \right.$$

$$\text{let } I = \int e^{3x} \sin x$$

So that:

$$I = -e^{3x} \cos x + 3 \left[e^{3x} \sin x - 3I \right] + C$$

$$I = -e^{3x} \cos x + 3e^{3x} \sin x - 9I + C$$

$$10I = e^{3x} (3 \sin x - \cos x) + C$$

$$\therefore I = \int e^{3x} \sin x \, dx = \frac{1}{10} e^{3x} (3 \sin x - \cos x) + C'$$

Example 2

Evaluate the integral: $\int_0^1 x \tan^{-1} x \, dx$

Solution

$$\int_0^1 x \tan^{-1} x \, dx = \frac{1}{2} x^2 \tan^{-1} x \Big|_0^1 - \frac{1}{2} \int_0^1 \frac{x^2}{1+x^2} \, dx$$

$$\text{but } \frac{x^2}{1+x^2} = 1 - \frac{1}{1+x^2}$$

$$\left\{ \begin{array}{l} \text{let } u = \tan^{-1} x, \quad v' = x \\ u' = \frac{1}{1+x^2}, \quad v = \frac{1}{2} x^2 \end{array} \right.$$

So that

$$\int_0^1 x \tan^{-1} x \, dx = \frac{1}{2} \tan^{-1}(1) - \frac{1}{2} \left[x \Big|_0^1 - \tan^{-1} x \Big|_0^1 \right]$$

$$= \frac{1}{2} \left(\frac{\pi}{4} \right) - \frac{1}{2} + \frac{1}{2} \left(\frac{\pi}{4} \right)$$

$$= \frac{\pi}{4} - \frac{1}{2}$$